

### Task 1 — Draw a Scatter Plot with Different Shapes

Create a scatter plot for:

x = [5,10,15,20,25]

y = [10,30,20,40,60]

#### Conditions:

- Each point must have **different color**
  - Each point must have **different size**
  - Add **grid, title, labels**
  - Add a **dotted line** (--) joining the points
- 

### Task 2 — Two Groups, Two Scatter Plots

Make 2 groups:

#### Group A

x = [1,3,5,7]

y = [10,18,12,25]

#### Group B

x = [2,4,6,8]

y = [22,28,30,35]

#### Conditions:

- Plot both groups using scatter
  - Give different colors
  - Add legend
  - Add title: "**Performance Comparison**"
  - Add x label: "**Time**"
  - Add y label: "**Score**"
- 

### Task 3 — Horizontal Bar Chart

Categories:

['Apple','Banana','Mango','Orange']

Values:

[50, 80, 40, 90]

**Conditions:**

- Use **different colors**
  - Show **value on each bar** using plt.text()
  - Add title: "**Fruit Sales Report**"
- 

#### **Task 4 — Histogram of Student Marks**

Create 100 random student marks between 1–100:

```
marks = np.random.randint(1,100,size=100)
```

**Conditions:**

- Show histogram with 10 bins
  - Color = blue
  - Edge color = black
  - Add title: "**Marks Distribution**"
- 

#### **Task 5 — Mixed Chart (Line + Scatter)**

Data:

```
x = [1,2,3,4,5]
```

```
y = [5,12,18,25,40]
```

**Conditions:**

- Draw line plot
  - Draw scatter plot on same graph
  - Line color = green
  - Scatter color = red
  - Add title, labels, grid
- 

#### **Task 6 — Compare 3 Bar Charts in One Graph**

Categories:

```
['Math','Science','English']
```

Scores:

```
student1 = [80,60,75]
```

```
student2 = [85,70,90]
```

```
student3 = [78,68,82]
```

**Conditions:**

- Use **bar chart** for 3 students in same graph
  - Different colors
  - Add legend
  - Add title: "**Student Performance Comparison**"
- 

 **Task 7 — Multiple Histograms**

Create 2 datasets:

```
d1 = np.random.randint(1,100,50)
```

```
d2 = np.random.randint(1,100,50)
```

**Conditions:**

- Plot both histograms on same graph
- Make them slightly transparent using **alpha**
- Give different colors
- Add legend
- Add title